$$
\begin{aligned}
|\mathbf{y}_i \cdot \mathbf{y}_j| &= \left| \left(\sum_{k=1}^{n_i} z_{i,k} \mathbf{v}_k \right) \cdot \left(\sum_{k=1}^{n_j} z_{j,k} \mathbf{u}_k \right) \right| \\
&= \left| \sum_{k_1=1}^{n_i} \sum_{k_2=1}^{n_j} (z_{i,k_1} \mathbf{v}_{k_1}) \cdot (z_{j,k_2} \mathbf{u}_{k_2}) \right| \\
&\leq \sum_{k_1=1}^{n_i} \sum_{k_2=1}^{n_j} |z_{i,k_1} z_{j,k_2}| \, |\mathbf{v}_{k_1} \cdot \mathbf{u}_{k_2}| && \text{Triangle Inequality} \\
&\leq \sum_{k_1=1}^{n_i} \sum_{k_2=1}^{n_j} |z_{i,k_1} z_{j,k_2}| \, \delta && \text{All } \mathbf{v}_i, \mathbf{u}_j \text{ are } \delta \text{ orthogonal} \\
&= \delta \sum_{k_1=1}^{n_i} \sum_{k_2=1}^{n_j} |z_{i,k_1} z_{j,k_2}| \\
&= \delta \left| \sum_{k=1}^{n_i} z_{i,k} \right| \left| \sum_{k=1}^{n_j} z_{j,k} \right| && \text{Factoring the product} \\
&\leq \delta \sqrt{\frac{n_i}{1 - \delta n_i}} \sqrt{\frac{n_j}{1 - \delta n_j}} && \text{By Lemma 1} \\
&\leq \frac{\delta d_{\max}}{1 - \delta d_{\max}} && n_i, n_j \leq d_{\max} \text{ by assumption}
\end{aligned}
$$

Thus $\mathbf{A}_i$ and $\mathbf{A}_j$ are $f(\delta)$-orthogonal for $f(\delta) = \delta d_{\max}/(1 - \delta d_{\max})$, and so it is possible to choose $\frac{1}{d_{\max}} e^{Cd\delta^2}$ pairwise $f(\delta)$-orthogonal projection matrices. Remapping the variable δ with $\delta \mapsto f^{-1}(\delta) = \delta/(d_{\max}(1 + \delta))$, we find that it is possible to choose $\frac{1}{d_{\max}} e^{Cd\delta^2/((1+\delta)^2 d_{\max}^2)}$ pairwise δ-orthogonal projection matrices. Because $1 + \delta$ is at most 2 with $\delta \in (0, 1)$, we can further simplify the exponent and find that it is possible to choose $\frac{1}{d_{\max}} e^{C(d/d_{\max}^2)\delta^2/4}$ pairwise δ-orthogonal projection matrices. Absorbing the 4 into the constant C finishes the proof of the lower bound.

For the upper bound, we can proceed much more simply. Consider k pairwise δ-orthogonal matrices $A_i \in \mathbb{R}^{d'}$. Since these matrices are full rank, their column spaces each parameterize a subspace of dimension d', and so by a result from (Alon, 2003) it is possible to choose $e^{Cd'\delta^2 \log(\frac{1}{\delta})}$ almost orthogonal vectors in this subspace. Furthermore, by our definition of δ-orthogonal matrices, all pairs of these vectors between subspaces will be δ-orthogonal. Finally, again by (Alon, 2003) we cannot have more than $e^{Cd\delta^2 \log(\frac{1}{\delta})}$ δ-orthogonal vectors overall, so we have that

$$
k e^{Cd_{\max}\delta^2 \log(\frac{1}{\delta})} < e^{Cd\delta^2 \log(\frac{1}{\delta})}
$$

and simplfying,

$$
k < e^{C(d - d_{\max})\delta^2 \log(\frac{1}{\delta})}
$$

□

These results imply that models can still represent an exponential number of higher dimensional features. However, there is a large exponential gap between the lower and upper bound we have shown. If the lower bound is reasonably tight, then this would mean that models would be highly incentivized to fit features within the smallest dimensional space possible, suggesting a reason for recent work showing interesting compressed encodings of multi-dimensional features in toy problems (Morwani et al., 2023).

Note that the proof assumes the "worst case" scenario that all of the features are dimension $d_{\max}$, while in practice many of the features may be 1 or low dimensional, so the effect on the capacity of a real model that represents multi-dimensional features is unlikely to be this extreme.

Finally, we note that the dictionary learning literature may have discovered similar results in the past (which we were unaware of), see Foucart & Rauhut (2013).

B More on Reducibility

B.1 Additional Intuition for Definitions

Here, we present some extra intuition and high level ideas for understanding our definitions and the motivation behind them. Roughly, we intend for our definitions in the main text to identify representations in the model that describe an object or concept in a way that fundamentally takes multiple dimensions. We operationalize this as finding a subspace of representations that 1. has basis vectors that "always co-occur" no matter the orientation 2. is not made up of combinations of independent lower-dimensional features.

1. The first condition is met by the mixture part of our definition. The feature in question should be part of an irreducible manifold, and so should "fill" a plane or hyperplane. There shouldn't be any part of the plane where the probability distribution of the feature is concentrated, because this region is then likely part of a lower dimensional feature. The idea of this part of the definition is to capture multi-dimensional objects; if the entire multi-dimensional space is truly being used to represent a high-dimensional object, then the representations for the object should be "spread out" entirely through the space.

2. The second condition is met by the separability part of our definition. This part of the definition is intended to rule out features that co-occur frequently but are fundamentally not describing the same object or concept. For example, latitude and longitude are not a mixture in that they frequently co-occur, but we do not think it is necessarily correct to say they are part of the same multi-dimensional feature because they are independent.

B.2 Empirical Irreducible Feature Test Details

Our tests for reducibility require the computation of two quantities $S(\mathbf{f})$ for the separability index and $M_\epsilon(\mathbf{f})$ for the ϵ-mixture index. We describe how we compute each index in the following two subsections.

B.2.1 Separability Index

We define the separability index in Equation 2 as

$$S(\mathbf{f}) = \min I(\mathbf{a}; \mathbf{b})$$

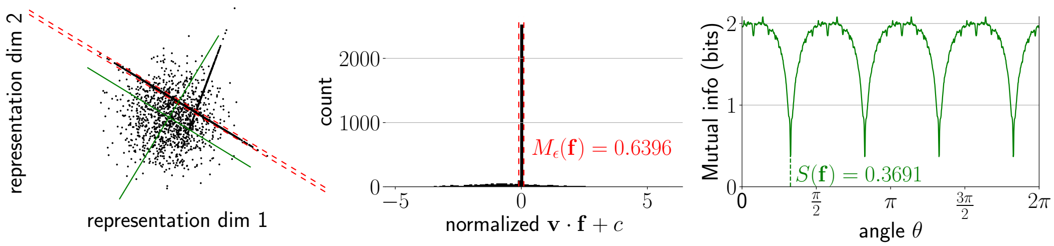
where the min is over rotations $\mathbf{R}$ used to split $\mathbf{f}' = \mathbf{R}\mathbf{f} + \mathbf{c}$ into $\mathbf{a}$ and $\mathbf{b}$. In two dimensions, the rotation is defined by a single angle, so we can iterate over a grid of 1000 angles and estimate the mutual information between $\mathbf{a}$ and $\mathbf{b}$ for each angle. We first normalize $\mathbf{f}$ by subtracting off the mean and then dividing by the root mean squared norm of $\mathbf{f}$ (and multiplying by $\sqrt{2}$ since the toy datasets are in two dimensions). To estimate the mutual information, we first clip the data $\mathbf{f}$ to a 6 by 6 square centered on the origin. We then bin the points into a 40 by 40 grid, to produce a discrete distribution $p(a, b)$. After computing the marginals $p(a)$ and $p(b)$ by summing the distribution over each axis, we obtain the mutual information via the formula

$$I(\mathbf{a}; \mathbf{b}) = \sum_{a,b} p(a, b) \log \frac{p(a, b)}{p(a)p(b)} \tag{8}$$

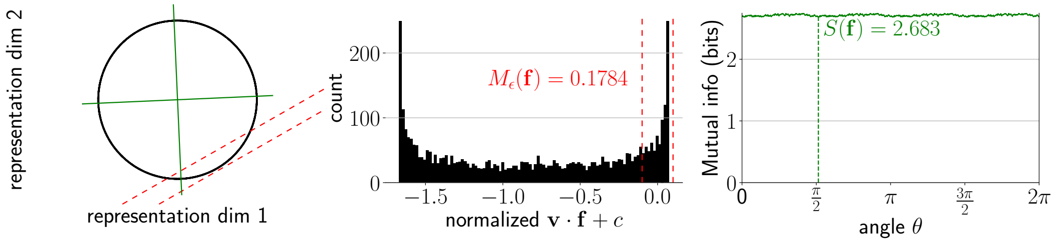
B.2.2 ϵ-Mixture Index

We define the ϵ-mixture index in Equation 3 as

$$M_\epsilon(\mathbf{f}) = \max_{\mathbf{v} \in \mathbb{R}^{d_f},\, c \in \mathbb{R}} \mathbb{P}\left(|\mathbf{v} \cdot \mathbf{f} + c| < \epsilon \sqrt{\mathbb{E}[(\mathbf{v} \cdot \mathbf{f} + c)^2]} \right)$$



(a) Testing $S(\mathbf{a})$ and $M_\epsilon(\mathbf{a})$ on a reducible feature $\mathbf{a}$.



(b) Testing $S(\mathbf{b})$ and $M_\epsilon(\mathbf{b})$ on an irreducible feature $\mathbf{b}$

Figure 10: Testing irreducibility of synthetic features. **Left in each subfigure:** Distributions of $\mathbf{x}$. For feature $\mathbf{a}$, 63.96% lies within the narrow dotted lines, indicating the feature is likely a mixture. For feature $\mathbf{b}$, 17.84% lies within the wide lines, indicating the feature is unlikely to be a mixture. The green cross indicates the angle θ that minimizes mutual information. **Middle in each subfigure:** Histograms of the distribution of $\mathbf{v} \cdot \mathbf{x}$ with red lines indicating a 2ϵ-wide region. **Right in each subfigure:** Mutual information between $\mathbf{a}$ and $\mathbf{b}$ as a function of the rotation angle θ of matrix $\mathbf{R}$. Feature $\mathbf{b}$ has a large minimum mutual information so is unlikely to be separable; feature $\mathbf{a}$ has a medium value of minimum mutual information of about 0.37 bits.

The challenge with computing $M_\epsilon(\mathbf{f})$ is to compute the maximum. We opted to maximize via gradient descent; and we guaranteed differentiability by softening the inequality $<$ with a sigmoid,

$$M_{\epsilon,T}(\mathbf{f}, \mathbf{v}, c) = \mathbb{E}\left(\sigma\left(\frac{1}{T}\left(\epsilon - \frac{|\mathbf{v} \cdot \mathbf{f} + c|}{\sqrt{\mathbb{E}[(\mathbf{v} \cdot \mathbf{f} + c)^2]}}\right)\right)\right) \tag{9}$$

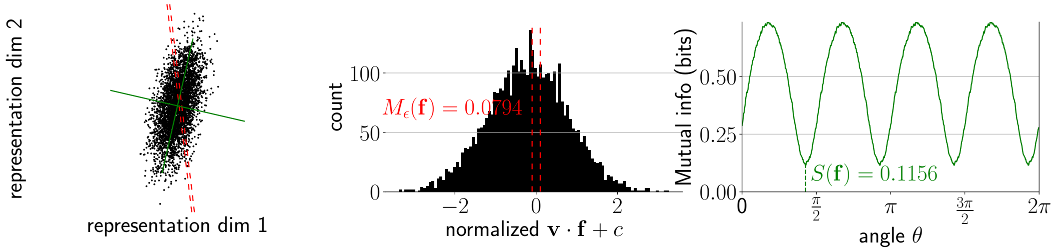
where T is a temperature, which we linearly decay from 1 to 0 throughout training. We optimize for $\mathbf{v}$ and c using this loss $M_{\epsilon,T}(\mathbf{f}, \mathbf{v}, c)$ using full batch gradient descent over 10000 steps with learning rate 0.1. With the solution $(\mathbf{v}^*, c^*)$, the final value of $M_{\epsilon,T=0}(\mathbf{f}, \mathbf{v}^*, c^*)$ is then our estimate of $M_\epsilon(\mathbf{f})$.

We also run the irreducibility tests on additional synthetic feature distributions in Fig. 11a and Fig. 11b.

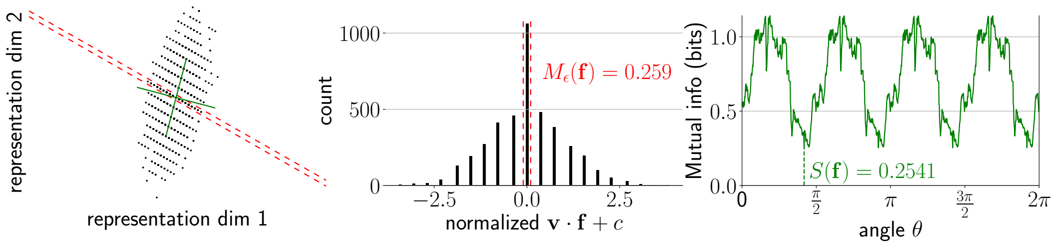
C Alternative Definitions

In this section, we present an alternative definition of a reducible feature that we considered during our work. This chiefly deals with multi-dimensional features from the angle of *computational* reducibility as opposed to *statistical* reducibility. In other words, this definition considers whether representations of features on a specific set of tasks can be split up without changing the accuracy of the task. This captures an interesting (and important) aspect of feature reducibility, but because it requires a specific set of prompts (as opposed to allowing unsupervised discovery) we chose not to use it as our main definition.

Our alternative definitions consider *representation spaces* that are possibly multi-dimensional, and defines these spaces through whether they can completely explain a function h on the output logits. We consider a *group theoretic* approach to irreducible representations, via whether computation involving multiple group elements can be decomposed.



(a) Testing $S(\mathbf{c})$ and $M_\epsilon(\mathbf{c})$ on a reducible feature $\mathbf{c}$.



(b) Testing $S(\mathbf{d})$ and $M_\epsilon(\mathbf{d})$ on an irreducible feature $\mathbf{d}$

Figure 11: Testing irreducibility of synthetic features. **Left in each subfigure:** Distributions of $\mathbf{x}$. For feature $\mathbf{c}$, 7.94% lies within the narrow dotted lines, indicating the feature is unlikely to be a mixture. For feature $\mathbf{d}$, 25.90% lies within the wide lines, indicating the feature is likely a mixture. The green cross indicates the angle θ that minimizes mutual information. **Middle in each subfigure:** Histograms of the distribution of $\mathbf{v} \cdot \mathbf{x}$ with red lines indicating a 2ϵ-wide region. **Right in each subfigure:** Mutual information between $\mathbf{a}$ and $\mathbf{b}$ as a function of the rotation angle θ of matrix $\mathbf{R}$. Both features have a small (< 0.5 bits) minimum mutual information and so are likely separable.

C.1 Alternative Definition: Interventions and Representation Spaces

Assume that we restrict the input set of prompts $T = \{\mathbf{t}^j\}$ to some subset of prompts and that we have some evaluation function h that maps from the output logit distribution of M to a real number. For example, for the `Weekdays` problems, T is the set of 49 prompts and h could be the $\arg\max$ over the days of week logits. Abusing notation, we let M also be the function from the layer we are intervening on; this is always clear from context. Then we can define a representation space of $\mathbf{x}^j_{i,l}$ as a subspace in which interventions always work:

Definition 5 (Representation Space)**.** Given a prompt set $T = \{\mathbf{t}^j\}$, a rank-r dimensional *representation space* of intermediate value $\mathbf{x}^j_{i,l}$ is a rank r projection matrix P such that for all j, j', $h(M((I-P)\mathbf{x}^j_{i,l} + P\mathbf{x}^{j'}_{i,l})) = h(M(\mathbf{x}^{j'}_{i,l}))$.

Note that it immediately follows that the rank d dimensional matrix I_d is trivially a rank d representation space for all prompt sets T.

Definition 6 (Minimality)**.** A representation space P of rank r is *minimal* if there does not exist a lower rank representation space.

A minimal representation with rank > 1 is a *multi-dimensional representation*.

Definition 7 (Alternative Reducibility)**.** A representation space P of rank r is *reducible* if there are orthonormal representation spaces P_1 and P_2 (such that $P_1 + P_2 = P$, $P_1 P_2 = 0$) where

$$h(M(P_1\mathbf{x}^j_{i,l}) + M(P_2\mathbf{x}^j_{i,l})) = h(M(P_1\mathbf{x}^j_{i,l} + P_2\mathbf{x}^j_{i,l}))$$

for all j, j'.

Suppose T, h and M define the multiplication of two elements in a finite group G of order n. Then if we interpret the embedding vectors as the group representations, our definition of reducibility implies to the standard group-theoretical definition of irreducibility — specifically, reducibility into a tensor product representation.

D Toy Case of Training SAEs on Circles

To explore how SAEs behave when reconstructing irreducible features of dimension $d_f > 1$, we perform experiments with the following toy setup. Inspired by the circular representations of integers that networks learn when trained on modular addition (Nanda et al., 2023a; Liu et al., 2022), we create synthetic datasets of activations containing multiple features which are each 2d irreducible circles.

First however, consider activations for a single circle – points uniformly distributed on the unit circle in $\mathbb{R}^2$. We train SAEs on this data with encoder $\texttt{Enc}(\mathbf{x}) = \texttt{ReLU}(\mathbf{W_e}(\mathbf{x} - \mathbf{b_d}) + \mathbf{b_e})$ and decoder $\texttt{Dec}(\mathbf{f}) = \mathbf{W_d}\mathbf{f} + \mathbf{b_d}$. We train SAEs with $m = 2$ and $m = 10$ with the Adam optimizer and a learning rate of 10^{-3}, sparsity penalty $\lambda = 0.1$, for 20,000 steps, and a warmup of 1000 steps. In Fig. 12 we show the dictionary elements of these SAEs. When $m = 2$, the SAE must use both SAE features on each input point, and uses $\mathbf{d_b}$ to shift the reconstructed circle so it is centered at the origin. When $m = 10$, the SAE learns $\mathbf{d_b} \approx 0$ and the features spread out across the circle, arranged close together, and only a subset are active on each input.

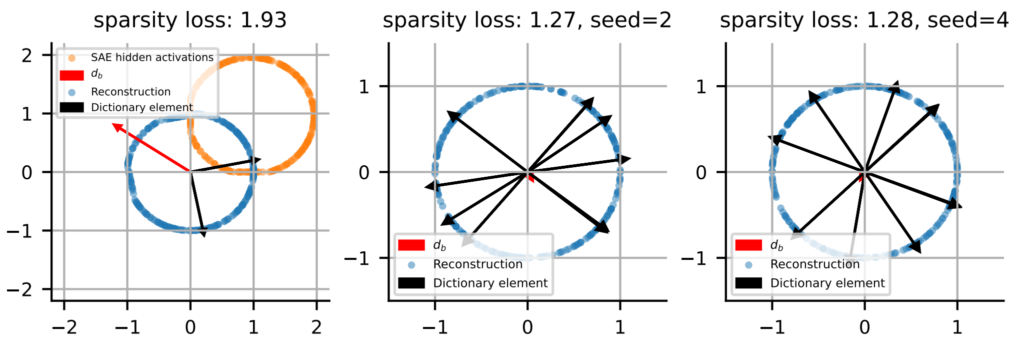


Figure 12: SAEs trained to reconstruct a single 2d circle with $m = 2$ (left) and $m = 10$ (middle and right) dictionary elements. When there are several SAE features, there is no natural/canonical choice of feature directions, and the dictionary elements spread out across the circle.

We now consider synthetic activations with multiple circular features. Our data consists of points in $\mathbb{R}^{10}$, where we choose two orthogonal planes spanned by $(\mathbf{e_1}, \mathbf{e_2})$ and $(\mathbf{e_3}, \mathbf{e_4})$, respectively. With probability one half a points is sampled uniformly on the unit circle in the $\mathbf{e_1}$-$\mathbf{e_2}$ plane, otherwise the point will be sampled uniformly on the unit circle in the $\mathbf{e_3}$-$\mathbf{e_4}$ plane. We train SAEs with $m = 64$ on this data with the same hyperparameters as the single-circle case.

We now apply the procedure described in Section 4 to see if we can automatically rediscover these circles. Encouragingly, we first find that the alive SAE features align almost exactly with either the $\mathbf{e_1}$-$\mathbf{e_2}$ or the $\mathbf{e_3}$-$\mathbf{e_4}$ plane. When we apply spectral clustering with $\texttt{n\_clusters} = 2$ to the features with the pairwise angular similarities between dictionary elements as the similarity matrix (Fig. 13, left), the two clusters correspond exactly to the features which span each plane. As described in Section 4, given a cluster of dictionary elements $S \subset \{1, \ldots, m\}$, we run a large set of activations through the SAE, then filter out samples which don't activate any element in S. For samples which do activate an element of S, reconstruct the activation while setting all SAE features not in S to have a hidden activation of zero. If some collection of SAE features together represent some irreducible feature, we want to remove all other features from the activation vector, and so we only allow SAE features in the collection to participate in reconstructing the input activation. We find that this procedure almost exactly recovers the original two circles, which encouraged us to apply this method for discovering the features shown in Fig. 1 and Fig. 15.

E Training Mistral SAEs

Our Mistral 7B (Jiang et al., 2023) sparse autoencoders (SAEs) are trained on over one billion tokens from a subset of the Pile (Gao et al., 2020) and Alpaca (Peng et al., 2023) datasets. We train our SAEs